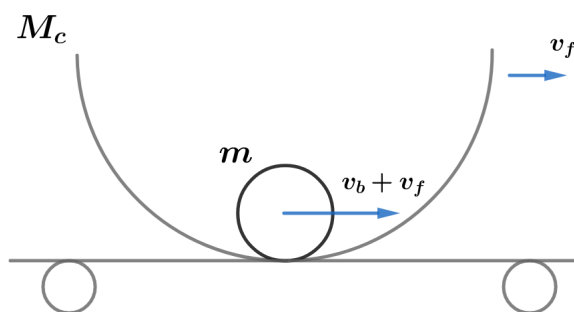


2019B F=ma Exam: Problem 10

Kevin S. Huang

Initially, the momentum of the system is

$$p_o = (m + M_C)v_0$$



When the ball is at the bottom of the bowl, the momentum of the system is

$$p_f = m(v_b + v_f) + M_cv_f$$

By conservation of momentum,

$$(m + M_c)v_0 = mv_b + (m + M_c)v_f$$

$$v_f = \frac{(m + M_c)v_0 - mv_b}{m + M_c}$$

so the answer is D.